

RELATIVE EXPANDERS AND OSTROVSKII'S BOUNDED-GEOMETRY QUESTION

RUN FA_BANACH_001

Status. `literature_implied_answer` (bounded-geometry version; group version remains open). This note records an existing literature answer, not a new theorem.

SOURCE QUESTION

In *Coarse embeddability into Banach spaces*, arXiv:0802.3666, Ostrovskii asks whether a metric space with bounded geometry that is not coarsely embeddable into Hilbert space must contain an expander-like substructure. Immediately afterward he quotes the more specific Gromov–Kasparov problem: if a finitely presented group has Cayley graph not coarsely embeddable into ℓ_2 , must that Cayley graph weakly contain a family of expanders?

SUPPORTING LITERATURE

Arzhantseva–Tessera, *Relative expanders*, arXiv:1402.1481, construct a finitely generated group G and finite-index normal subgroups (N_n) such that, for every finite generating set S , the finite Cayley graphs

$$(G/N_n, S)$$

do not coarsely embed into any L^p -space for $1 \leq p < \infty$, and more generally not into any uniformly curved Banach space, yet admit no weakly embedded expander.

IDENTIFICATION

The coarse disjoint union of the graphs $(G/N_n, S)$ is a bounded-geometry metric space. Since the sequence does not coarsely embed into any L^p , it does not coarsely embed into Hilbert space. Since the sequence admits no weakly embedded expander, it gives a negative answer to Ostrovskii's broader bounded-geometry question: non-embeddability into Hilbert does not force a weakly embedded expander.

SCOPE

This does not settle the narrower group formulation printed by Ostrovskii. Arzhantseva–Tessera themselves explicitly leave open the problem of constructing a finitely generated group whose Cayley graph does not coarsely embed into Hilbert space and yet has no weakly embedded expander. Ostrovskii’s quoted problem asks the even narrower finitely presented version. Thus this packet records only the bounded-geometry metric-space answer.

The separate question in Ostrovskii’s Section 5 asking whether ℓ_2 coarsely embeds into every infinite-dimensional Banach space is already covered in the run registry by the Baudier–Lancien–Schlumprecht answer, arXiv:1705.06797.

PROVENANCE

The supporting paper does not appear to cite arXiv:0802.3666 as the source of the bounded-geometry question; the relation is therefore recorded as a literature-implied answer. It is not a new counterexample from this run.